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Om Mehta

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1 Summarizing the logic

The paper first highlights the fact that 80% of US citizens live in large cities, compared with 58% in Europe. And since so much of the country's population is in large cities, it makes sense that a **large** share of GDP comes from those cities. In addition to large cities (like NY and LA, which are among the top cities in the world in terms of GDP), the 28 **middleweight** cities are responsible for 37% of US GDP. All the middleweight cities combined make up more than 70% of the US GDP.

The point that the paper is trying to make is that not only does the US have **many** middleweights that pull the nation's GDP higher, but those middleweights have very high GDP per capita. GDP per large/mid city is proportionate to its population. And, in comparison to other nations, like Europe, the dense populations make way for more competition, more innovation, and eventually more productivity.

I believe that the main point the paper is trying to convey is that in other countries like **Europe** and **Japan**, their big chunk of GDP comes from a few large cities (London, Paris, Tokyo, Osaka). Simply put, the US has more geographical opportunity for growth, across the entire nation, making way for less entry barriers into new markets and more stable economic growth. Diversification wins big in this sense over nations (like Japan) whose 2-3 biggest cities account for almost half their GDP.

2 Adding structure

Using the fact that **marginal product of labor in each city is equal to the wage in the city**, we can build the following equality:

$$\frac{\partial Y}{\partial L} = A(1 - \gamma)L^{-\gamma} = w$$

Where:

- Y = output (total amount of goods & services produced)
- L = labor (number of workers)
- A = total factor productivity
- γ = elasticity of output with respect to labor ("how much more productivity do we get for 1 more worker?")

Let w_L and w_M equal the wages of the large cities and middleweight cities, respectively.

$$\begin{aligned}w_L &= 1.35w_M \\A_L(1 - \gamma)L_L^{-\gamma} &= 1.35A_M(1 - \gamma)L_M^{-\gamma}\end{aligned}$$

We can plug in $L_L = 0.8$ and $L_M = 0.2$ in order to show that large cities account for 80 percent of total employment, and put in $\gamma = 0.5$ for simplification. We first cancel out the $(1 - \gamma)$ on both sides, and then continue:

$$\begin{aligned}A_L L_L^{-0.5} &= 1.35 A_M L_M^{-0.5} \\ \Rightarrow \frac{A_L}{A_M} &= \frac{1.35 L_M^{-0.5}}{L_L^{-0.5}} \\ &= \frac{1.35 / \sqrt{0.2}}{1 / \sqrt{0.8}} \\ \Rightarrow \frac{A_L}{A_M} &= 2.7\end{aligned}$$

This means that if large cities accounted for 80% of total employment in the US, the **larger cities are outputting 2.7x more per worker** than middleweight cities.

3 Employment share with identical wages

In order to keep the $\frac{A_L}{A_M} = 2.7$ productivity gap, while having the same wage in large and middleweight cities, the share of employment would have to decrease in large cities and increase in middleweights. The math to find this out is pretty straightforward. First, we reuse our equation from before but delete the 1.35 since we now have identical wages:

$$A_L L_L^{-\gamma} = A_M L_M^{-\gamma}$$

Using our ratio from earlier and exponential identity:

$$\left(\frac{L_M^{-\gamma}}{L_L^{-\gamma}} \right) = \left(\frac{L_L}{L_M} \right)^{\gamma} = 2.7$$

$$\gamma = 0.5 : \frac{L_L}{L_M} = \sqrt{2.7}$$

We can now solve this two variable system to get the share of labor:

$$\frac{L_L}{L_M} \approx 1.64$$

$$L_L + L_M = 1$$

$$L_L = 1 - L_M$$

$$\frac{1 - L_M}{L_M} = 1.64$$

$$1 - L_M = 1.64 L_M$$

$$2.64 L_M = 1$$

$$\boxed{L_M = 0.38, L_L = 0.62}$$

So, in order to maintain the productivity gap and make wages equal, there would have to be an 18% shift in total employment share.

4 Larger wage premium

In order to calculate the share of population (which is analogous to the share of employment in this example) L_L and L_M when $w_L = 4.83w_M$, while maintaining the same productivity ratio, we can take a similar approach as of that in question (3):

$$\begin{aligned}
A_L L_L^{-\gamma} &= 4.83 A_M L_M^{-\gamma} \\
2.7 &= 4.83 \left(\frac{L_L}{L_M} \right)^\gamma \\
\frac{L_L}{L_M} &= \sqrt[\gamma]{\frac{2.7}{4.83}} \approx 0.7477 \\
L_L + L_M &= 1 \\
\frac{1 - L_M}{L_M} &= 0.7477 \\
1 - L_M &= 0.7477 L_M \\
1.7477 L_M &= 1
\end{aligned}$$

$$L_M = 0.57, L_L = 0.43$$

If the wage gap was similar to China's, the share of employment in large US cities would have to be $\simeq 43\%$. This is a dramatic shift from earlier, where the majority of the workforce is now in middleweight cities.

5 International analysis

Based on my answers from (3) and (4), I gathered that, in order to maintain constant productivity, **employment share in large cities must decrease as wage premiums increase**. In simpler terms, as the wage gap grows, the employment gap narrows.

Using this result, we can deduce that in large cities across Europe, LATAM, and China, the share of the population is lower than in the U.S., because the wage premium is significantly higher. For instance, if wages in large Chinese cities are nearly four times higher than in the rest of the country, then only a relatively small portion of the workforce can be employed in those cities. Otherwise, if a large share of the population worked in these high-wage cities, the implied productivity would have to fall—mathematically—unless wages fell too.

This helps explain why the U.S. wage premium in large cities is only 35 percent: **the workforce is more diversified**. Whether due to geographic mobility, widespread infrastructure, or strong regional economies outside major

metros, U.S. labor is more evenly spread. This balance in urban employment may also help explain why internal social and economic divisions have remained manageable—contributing to the U.S.’s long-standing position as the world’s reserve currency.

6 Change in Top 30

We can see in the paper that many cities increased, decreased, or remained unchanged in terms of GDP growth. The goal here is to classify: among the that cities grew, which grew due to increasing population, and which grew by increasing productivity per individual?

Using the information in the paper, we can classify the changes in ranking of the Top 30 cities by Total GDP by those whose rankings increased, decreased, and remained unchanged:

Unchanged	Decreased	Increased
New York	Philadelphia	Washington, DC
Los Angeles	Detroit	Dallas
Chicago	San Francisco	Boston
Houston	San Diego	Atlanta
Miami	St. Louis	Phoenix
Seattle	Cleveland	Baltimore
Minneapolis	Pittsburgh	Denver
	Hartford	Riverside
		Tampa
		Portland (OR)

Cities Dropped from the Top 30 (1978 → 2010):

- New Orleans
- Milwaukee
- Columbus (OH)
- Hartford
- Buffalo

Cities Added to the Top 30 by 2010 (not in 1978):

- Riverside
- Tampa
- Portland (OR)
- Orlando

- Sacramento

Using this information, we can narrow down the countries to investigate to only the **increased** column, along with those cities added into the Top 30.

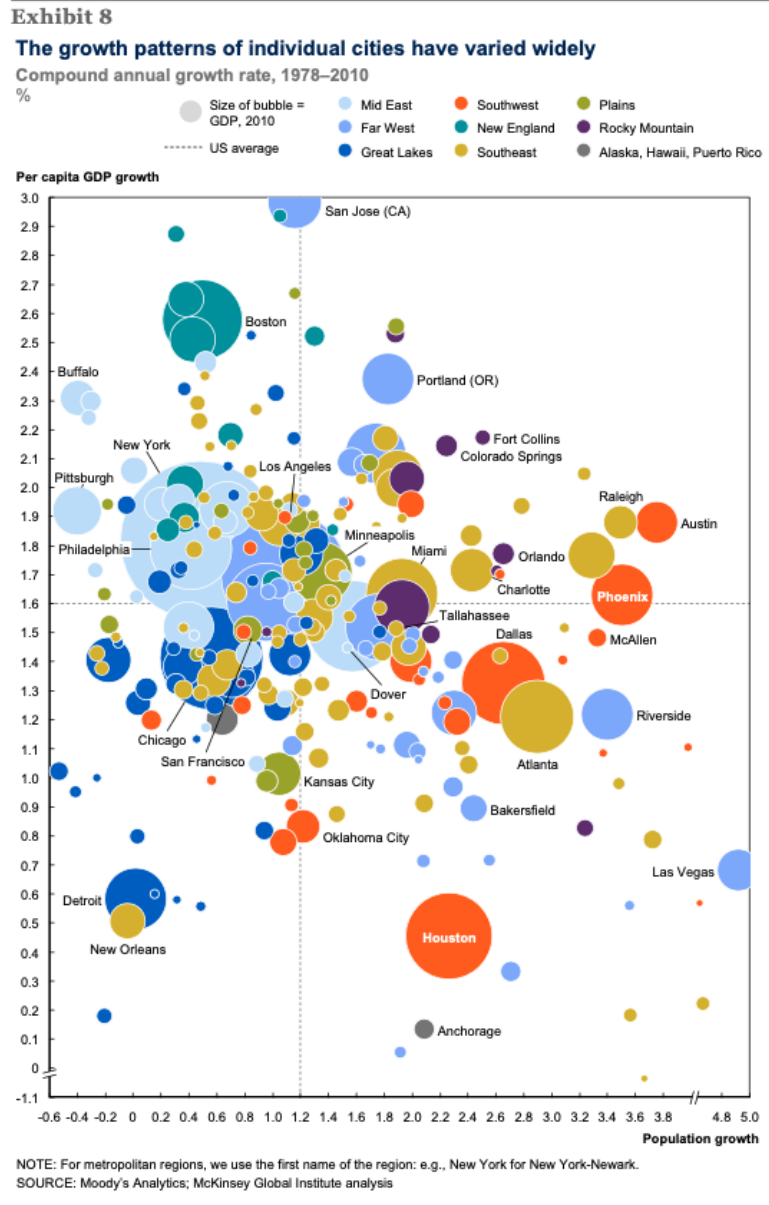


Figure 1: Growth–Population vs GDP per capita

We can see a very sharp distinction between cities in terms of where their growth came from, confirming the paper’s suggestion that cities can grow in either manner—increasing population or increasing GDP per capita. The first example is **Dallas** and **Boston**, where Dallas increased ranking primarily through an increase in population ($\sim 2.6x$), and Boston increased ranking primarily through Per capita GDP growth (also $\sim 2.6x$). This stark contrast helps prove the point because both cities increased in each metric by roughly the same amount. This is true for many of the other cities as well.